Find a Gene Assignment Part 1

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Q1. Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.
If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: Interleukin-36 gamma (IL36G)

Species: Homo sapiens

Accession: Q9NZH8

Function: Interleukin-36 gamma (IL36G) plays a role in skin inflammation and issume signaling. IL36G is associated in skin disorders such as psoriasis and dandruff.

Q2. Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN search against reptilia nt

Database: Nucleotide collection (nt)

Organsim: Reptilia (taxid:8459)

Also include the output of that BLAST search in your document. If appropriate, change the font to Courierdd size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press -shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.

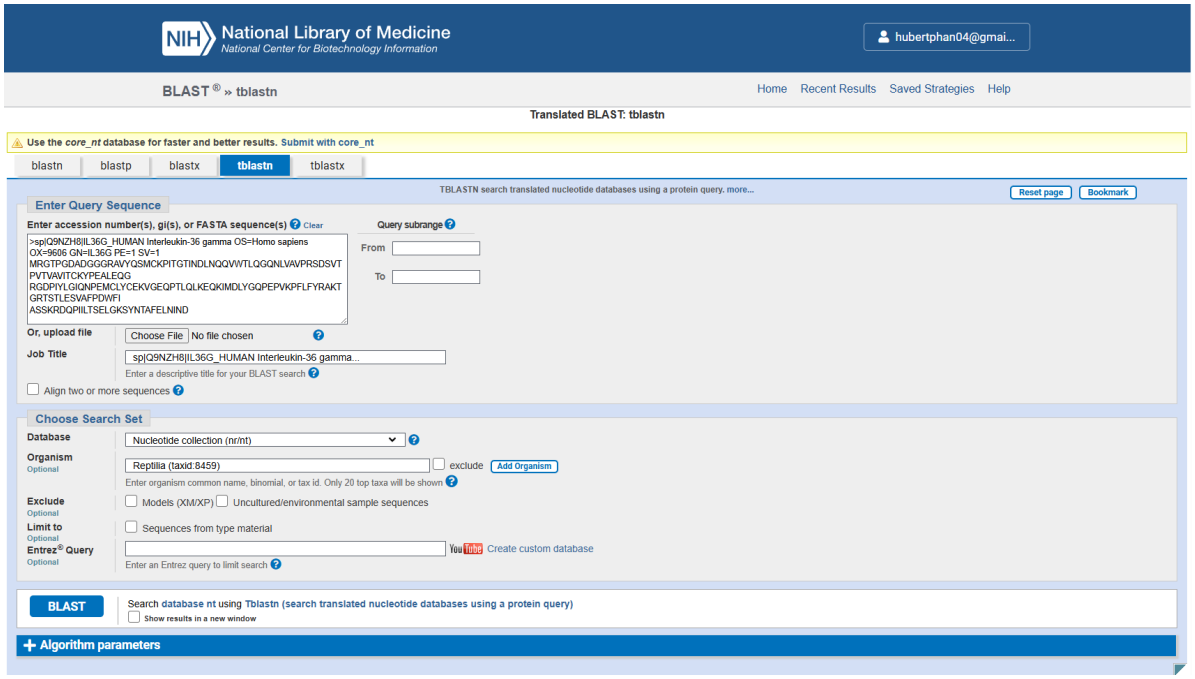

Figure 1: TBLASTN QUERY

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen Match: Accession OZ346534.1, Trionyx triunguis genome assembly. See alignment details below.

Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Your search is limited to records **include: Reptilia (taxid:8459)**

Job Title	**sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma...**
RID	RU2HMSBS016 *Search expires on 02-01 02:44 am* Download All
Program	TBLASTN Citation
Database	nt See details
Query ID	lcl|Query_6676515
Description	sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma OS=Hor ...
Molecule type	amino acid
Query Length	169
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter **Reset**

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *73 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819404), m...	Chelonoidis abi...	95.5	95.5	90%	1e-20	35.29%	3011	XM_032771100.2
PREDICTED: Emys orbicularis interleukin-36 receptor antagonist protein-like (LOC135873734), mRNA	Emys orbicularis	87.8	87.8	89%	1e-19	35.53%	576	XM_065398370.1
PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819405), m...	Chelonoidis abi...	87.8	87.8	89%	1e-19	34.87%	576	XM_032771101.1
PREDICTED: Gopherus flavomarginatus interleukin-36 receptor antagonist protein-like (LOC127044660)...	Gopherus flavo...	87.4	87.4	89%	2e-19	34.87%	576	XM_050939738.1
PREDICTED: Malaclemys terrapin pileata interleukin-36 receptor antagonist protein-like (LOC12883275...	Malaclemys terr...	90.1	90.1	85%	2e-19	36.81%	1090	XM_054020339.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	3e-19	37.06%	764	XM_039525790.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2253	XM_045006715.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2267	XM_045006714.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2207	XM_045006713.1
PREDICTED: Chrysemys picta bellii interleukin-36 receptor antagonist protein (LOC101949597), mRNA	Chrysemys pict...	90.1	180	85%	4e-19	36.81%	1357	XM_005279282.4
PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	90.1	90.1	85%	5e-19	36.81%	1389	XM_034759399.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	5e-19	37.06%	836	XM_039525788.1
PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371479...	Emydura macq...	87.8	87.8	96%	6e-19	32.75%	821	XM_067540356.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	6e-19	37.06%	896	XM_039525787.1
PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004348), m...	Carettochelys in...	85.5	85.5	91%	7e-19	34.19%	529	XM_074981942.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	88.2	88.2	82%	1e-18	36.69%	1072	XM_045007705.1
PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371478...	Emydura macq...	83.6	83.6	90%	2e-17	33.33%	787	XM_067540265.1
PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	82.4	82.4	85%	2e-16	34.84%	1434	XM_034759398.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	82%	4e-15	35.25%	1692	XM_075910708.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	79%	5e-15	35.82%	1690	XM_075910709.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	78.6	78.6	79%	6e-15	35.82%	1857	XM_075910706.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	75.5	75.5	53%	4e-14	43.96%	949	XM_045007706.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	75.1	75.1	90%	9e-14	31.37%	1468	XM_075910776.1
PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004315), m...	Carettochelys in...	72.4	72.4	85%	2e-13	32.88%	653	XM_074981886.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	73.6	73.6	90%	3e-13	31.37%	1428	XM_075910775.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.6	71.6	88%	1e-12	31.13%	1290	XM_075910855.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1290	XM_075910852.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1180	XM_075910854.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	70.9	70.9	88%	2e-12	31.13%	1283	XM_075910851.1

Figure 2: TBLASTN RESULTS

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Trionyx triunguis genome assembly, chromosome: 30

Sequence ID: OZ346534.1 Length: 17913073 Number of Matches: 1

Range 1: 1420032 to 1420238 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
50.1 bits(118)	5e-05	Compositional matrix adjust.	26/69(38%)	41/69(59%)	1/69(1%)	+3

```
Query  91       VGEQPTLQLKEQKIMDLYGQPEPVKPFLFYRAKTGRTSTLESVAFPDWFIASSKRD-QPI  149
                +   P    ++++IMDLY   +  K F F    TG T   ESVAFP WF+++S +D QP+
Sbjct  1420032  LSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQPL  1420211

Query  150      ILTSELGKS  158
                 +T+  G++
Sbjct  1420212  RVTNISGEA  1420238
```

Figure 3: Alignment Details

In general, [Q2] is the most difficult for students because it requires you to have a "feel" for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not "novel"), a near match (something that might be "novel", depending on the results of [Q4]), and a non-homologous result.
If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

Q3. Gather information about this "novel" protein. At a minimum, show me the protein sequence of the "novel" protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don't forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don't have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

The genomic DNA region surrounding the tblastn hit in Trionyx triunguis was extracted from the genome assembly and expanded beyond the aligned coordinates to capture potential coding sequence. The sequence was then translated in all six reading frames using EMBOSS Transeq. The longest open reading frame (ORF) without a stop codon was identified by examining for conserved motifs observed in the original BLAST alignment. This segment is the protein sequence used for downstream BLASTP analysis to assess sequence novelty.

Chosen Sequence:

```
>Trionyx_triunguis_tblastn_hit_translated_fragment frame_2
GQWADGRSLPPSLQAVSPFLIGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGW
FLSTSPKDGQPLRVTNISGEAEIINFYFKKM
```

> Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as S. cerevisiae, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: IL-1 family-like protein

Species: Trionyx triunguis (African softshell turtle)

Lineage: Eukaryota; Metazoa; Chordata; Reptilia; Testudines; Trionychidae; Trionyx

> **Q4.** For the purposes of this project, "novel" is defined as follows. Take the protein sequence (your answer to [Q3]) and use it as a query in a BLASTP search of the nr database at NCBI.

- If there is a match with **100% amino acid identity** to a protein in the database **from the same species**, then your protein is **NOT novel** (even if the match is to a protein with a name such as "unknown"). Someone has already found and annotated this sequence and assigned it an accession number.
- If the **top match reported has less than 100% identity**, then it is likely that your protein is **novel**, and you have succeeded.
- If there is a match with **100% identity**, but to a **different species** than the one you started with, then you have likely succeeded in finding a novel gene.
- If there are **no database matches** to the original query from [Q1], this indicates that you have **partially succeeded**: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

Details:

A BLASTP search against the nr database yielded a top hit protein from *Pelodiscus sinensis* (Chinese softshell turtle).

Below are screenshots of the BLASTP query, BLASTP results, and alignment details of the top hit.

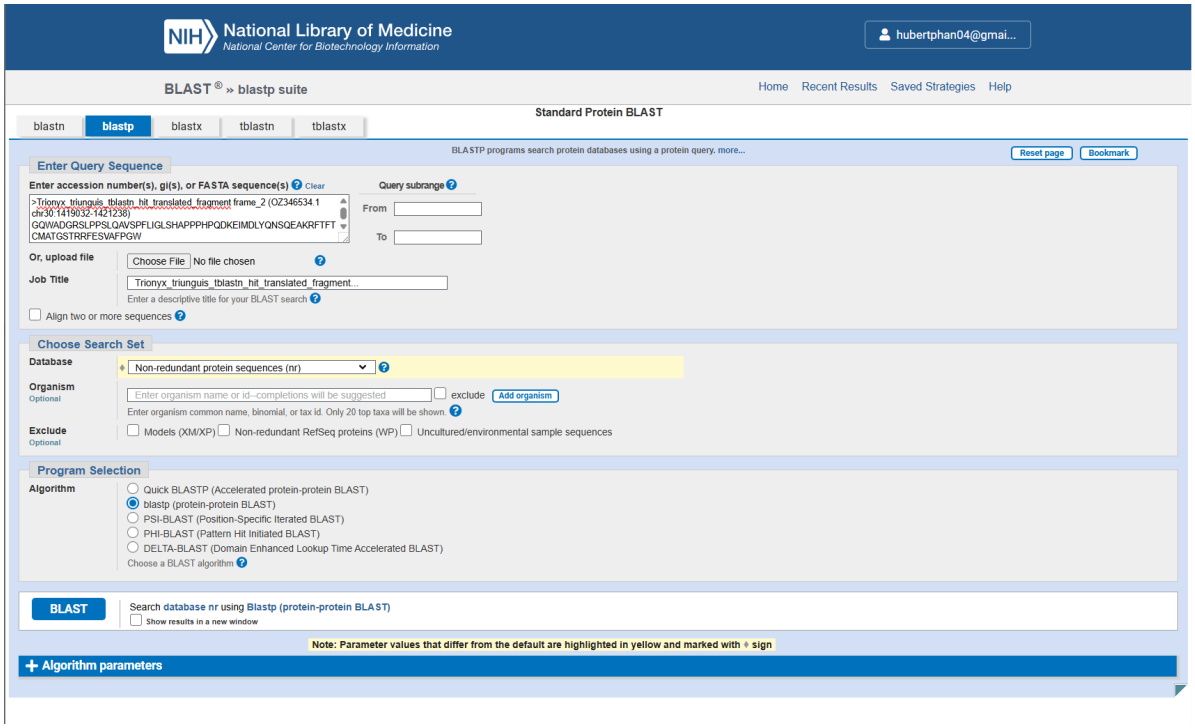
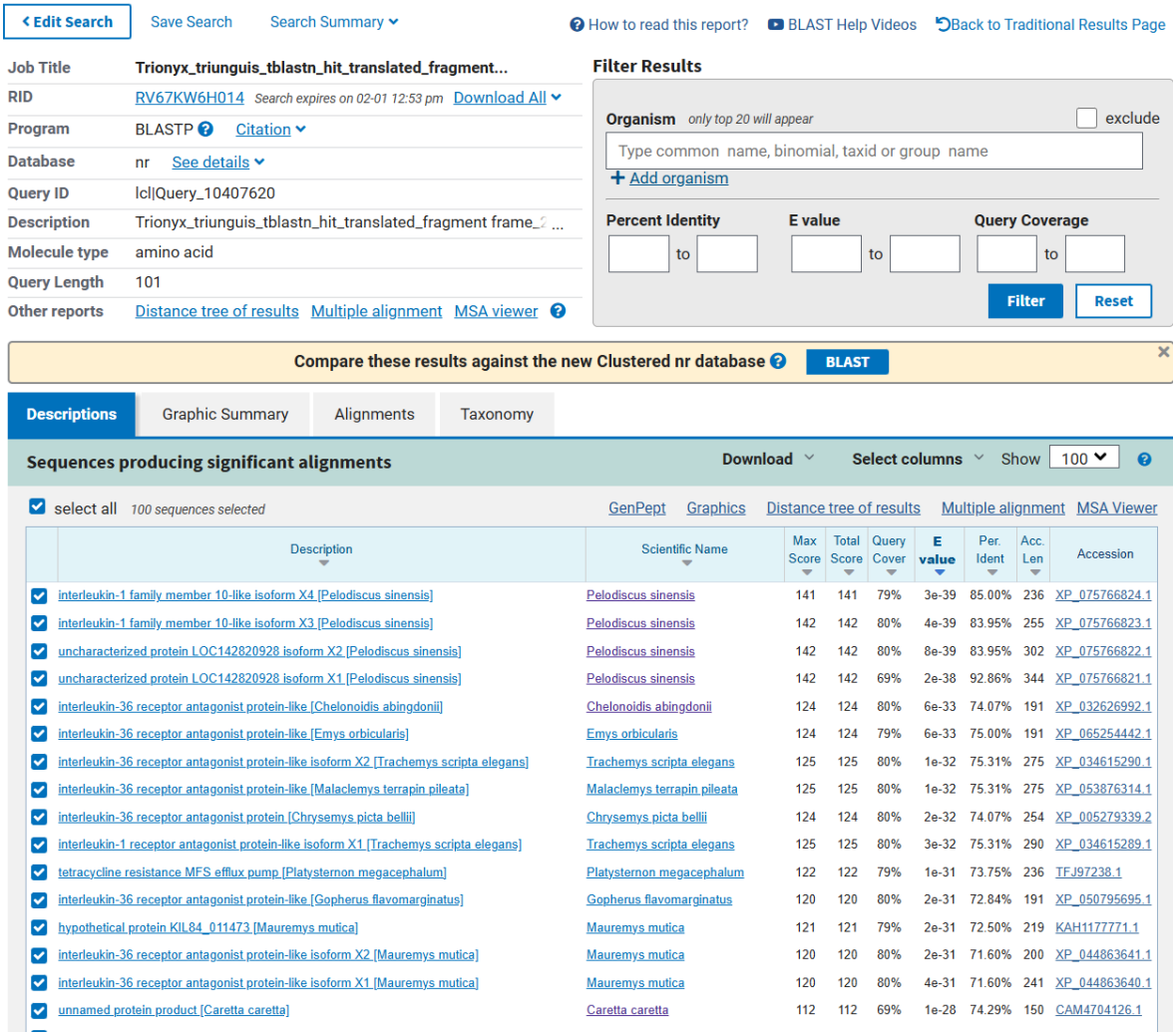


Figure 4: BLASTP Query



< Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Job Title	Trionyx_triunguis_tblastn_hit_translated_fragment...
RID	RV67KW6H014 Search expires on 02-01 12:53 pm Download All
Program	BLASTP Citation
Database	nr See details
Query ID	lcl|Query_10407620
Description	Trionyx_triunguis_tblastn_hit_translated_fragment frame_2 ...
Molecule type	amino acid
Query Length	101
Other reports	Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism only top 20 will appear exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to | E value to | Query Coverage to

Filter Reset

Compare these results against the new Clustered nr database BLAST

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all 100 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
interleukin-1 family member 10-like isoform X4 [Pelodiscus sinensis]	Pelodiscus sinensis	141	141	79%	3e-39	85.00%	236	XP_075766824.1
interleukin-1 family member 10-like isoform X3 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	80%	4e-39	83.95%	255	XP_075766823.1
uncharacterized protein LOC142820928 isoform X2 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	80%	8e-39	83.95%	302	XP_075766822.1
uncharacterized protein LOC142820928 isoform X1 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	69%	2e-38	92.86%	344	XP_075766821.1
interleukin-36 receptor antagonist protein-like [Chelonoidis abingdonii]	Chelonoidis abingdonii	124	124	80%	6e-33	74.07%	191	XP_032626992.1
interleukin-36 receptor antagonist protein-like [Emys orbicularis]	Emys orbicularis	124	124	79%	6e-33	75.00%	191	XP_065254442.1
interleukin-36 receptor antagonist protein-like isoform X2 [Trachemys scripta elegans]	Trachemys scripta elegans	125	125	80%	1e-32	75.31%	275	XP_034615290.1
interleukin-36 receptor antagonist protein-like [Malaclemys terrapin pileata]	Malaclemys terrapin pileata	125	125	80%	1e-32	75.31%	275	XP_053876314.1
interleukin-36 receptor antagonist protein [Chrysemys picta bellii]	Chrysemys picta bellii	124	124	80%	2e-32	74.07%	254	XP_005279339.2
interleukin-1 receptor antagonist protein-like isoform X1 [Trachemys scripta elegans]	Trachemys scripta elegans	125	125	80%	3e-32	75.31%	290	XP_034615289.1
tetracycline resistance MFS efflux pump [Platysternon megacephalum]	Platysternon megacephalum	122	122	79%	1e-31	73.75%	236	TFJ97238.1
interleukin-36 receptor antagonist protein-like [Gopherus flavomarginatus]	Gopherus flavomarginatus	120	120	80%	2e-31	72.84%	191	XP_050795695.1
hypothetical protein KIL84_011473 [Mauremys mutica]	Mauremys mutica	121	121	79%	2e-31	72.50%	219	KAH1177771.1
interleukin-36 receptor antagonist protein-like isoform X2 [Mauremys mutica]	Mauremys mutica	120	120	80%	2e-31	71.60%	200	XP_044863641.1
interleukin-36 receptor antagonist protein-like isoform X1 [Mauremys mutica]	Mauremys mutica	120	120	80%	4e-31	71.60%	241	XP_044863640.1
unnamed protein product [Caretta caretta]	Caretta caretta	112	112	69%	1e-28	74.29%	150	CAM4704126.1

Figure 5: BLASTP Results

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interleukin-1 family member 10-like isoform X4 [Pelodiscus sinensis]

Sequence ID: XP_075766824.1 Length: 236 Number of Matches: 1

Range 1: 157 to 236 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
141 bits(356)	3e-39	Compositional matrix adjust.	68/80(85%)	72/80(90%)	0/80(0%)

```
Query  22   GLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQP  81
            G S  P    +DKEIM LYQNSQEAKRFTFTC+ATGSTRRFESVAFPGWFLSTSPKDGQP
Sbjct  157  GTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTSPKDGQP  216

Query  82   LRVTNISGEAEIINFYFKKM  101
            +RVTNISGEAEI+NFYFKKM
Sbjct  217  VRVTNISGEAEIVNFYFKKM  236
```

Related Information

Genome Data Viewer - aligned genomic context

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interleukin-1 family member 10-like isoform X3 [Pelodiscus sinensis]

Sequence ID: XP_075766823.1 Length: 255 Number of Matches: 1

Range 1: 175 to 255 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
142 bits(357)	4e-39	Compositional matrix adjust.	68/81(84%)	72/81(88%)	0/81(0%)

```
Query  21   IGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQ  80
             G S  P    +DKEIM LYQNSQEAKRFTFTC+ATGSTRRFESVAFPGWFLSTSPKDGQ
Sbjct  175  CGTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTSPKDGQ  234

Query  81   PLRVTNISGEAEIINFYFKKM  101
            P+RVTNISGEAEI+NFYFKKM
Sbjct  235  PVRVTNISGEAEIVNFYFKKM  255
```

Related Information

Genome Data Viewer - aligned genomic context

Figure 6: Alignment Details